

GDOV, Russian Federation

WMO: 261570

Lat: 58.733N

Lon: 27.833E

Elev: 41

StdP: 100.83

Time Zone: 3.00 (E03)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.7	-19.7	-25.0	0.4	-22.6	-22.2	0.5	-19.2	9.5	0.2	8.8	-1.1	1.3	40	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	27.9	20.1	26.0	19.3	24.2	18.4	21.3	26.0	20.2	24.4	19.1	23.2	3.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	14.4	23.7	18.5	13.4	22.8	17.5	12.6	21.7	61.9	26.5	58.1	24.2	54.5	23.2	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.4	6.7	DB	-25.7	30.3	4.4	1.9	-28.9	31.7	-31.5	32.7	-33.9	33.8	-37.1	35.1	(4)
(5)			WB	-25.9	22.6	4.3	1.2	-29.0	23.5	-31.6	24.2	-34.0	24.8	-37.1	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-6.7	-5.2	-1.6	5.3	11.9	15.3	18.6	16.6	12.6	5.5	1.9	-1.8	(6)	
	DBStd	9.54	6.62	6.68	4.61	3.65	4.20	3.35	3.31	3.09	3.08	3.68	4.13	5.00	(7)	
	HDD10.0	2243	517	425	358	148	27	2	0	1	12	144	242	367	(8)	
	HDD18.3	4543	776	658	617	391	205	101	38	70	174	397	492	625	(9)	
	CDD10.0	818	0	0	0	7	85	162	265	206	88	5	0	0	(10)	
	CDD18.3	78	0	0	0	0	4	11	44	17	1	0	0	0	(11)	
	CDH23.3	564	0	0	0	0	44	85	314	118	3	0	0	0	(12)	
	CDH26.7	107	0	0	0	0	6	9	64	28	0	0	0	0	(13)	
(14) Wind	WSAvg	3.3	3.5	3.4	3.5	3.1	3.0	3.1	2.8	2.9	3.2	3.4	3.8	4.1	(14)	
(15) Precipitation	PrecAvg	653	45	35	33	33	51	78	71	84	57	64	57	45	(15)	
	PrecMax	839	83	62	64	76	106	141	154	230	118	109	103	88	(16)	
	PrecMin	465	18	9	9	8	12	11	9	0	10	16	5	20	(17)	
	PrecStd	99	16	14	16	18	26	35	35	54	27	22	22	18	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	5.0	10.8	20.5	27.2	27.8	30.5	29.7	23.4	14.6	9.8	6.8	(19)	
		MCWB	4.2	3.2	4.1	11.4	17.1	19.7	21.6	20.4	18.7	12.1	8.5	5.8	(20)	
	2%	DB	3.5	3.2	7.4	17.1	24.1	25.9	28.6	26.6	20.7	13.0	8.2	5.6	(21)	
		MCWB	2.8	1.9	3.9	9.6	16.3	18.2	20.8	20.0	16.2	11.2	7.4	4.8	(22)	
	5%	DB	2.1	2.3	5.4	14.4	21.6	23.7	26.6	24.1	18.7	11.8	7.4	4.5	(23)	
		MCWB	1.6	1.4	2.9	8.2	14.7	17.5	20.3	18.3	15.3	10.4	6.8	3.8	(24)	
	10%	DB	1.1	1.6	3.9	11.9	19.6	21.7	24.9	22.2	17.2	10.8	6.4	3.6	(25)	
		MCWB	0.7	0.9	2.0	6.6	13.3	16.0	19.3	17.6	14.2	9.5	5.9	2.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	3.5	5.7	12.0	19.4	21.1	23.2	22.3	18.5	12.9	8.9	6.2	(27)	
		MCDB	4.7	4.6	8.0	18.0	26.0	25.7	27.5	27.7	22.4	14.0	9.7	6.6	(28)	
	2%	WB	2.8	2.2	4.3	10.2	17.4	19.6	21.9	20.3	17.1	11.7	7.6	4.9	(29)	
		MCDB	3.3	2.8	6.5	16.1	22.1	23.9	27.3	24.3	19.9	12.7	8.0	5.4	(30)	
	5%	WB	1.7	1.6	3.3	8.9	15.8	18.2	20.9	19.0	15.8	10.8	6.9	3.9	(31)	
		MCDB	2.1	2.2	5.0	13.6	20.3	22.6	25.3	23.3	17.8	11.6	7.4	4.5	(32)	
	10%	WB	0.7	1.1	2.3	7.4	14.3	16.8	19.9	18.0	14.7	9.7	5.9	3.0	(33)	
		MCDB	1.0	1.6	3.8	11.0	18.6	20.6	23.8	21.6	16.6	10.6	6.3	3.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.1	6.6	8.8	10.0	8.6	8.4	8.2	6.6	5.2	3.1	3.2	(35)	
		MCDBR	3.1	3.6	7.2	14.0	12.9	11.5	11.1	11.8	9.1	5.7	3.1	3.3	(36)	
		MCWBR	2.9	3.1	5.0	7.6	6.4	6.0	5.0	5.8	5.4	4.6	2.9	3.3	(37)	
	5% WB	MCDBR	3.1	3.3	6.3	12.2	10.8	10.1	9.6	10.3	7.9	5.4	3.0	3.2	(38)	
		MCWBR	3.0	3.0	4.8	7.1	6.0	6.1	4.9	5.8	5.0	4.6	3.0	3.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.276	0.309	0.337	0.376	0.370	0.367	0.390	0.389	0.360	0.336	0.311	0.266	(40)	
	taud		2.112	2.131	2.205	2.306	2.395	2.436	2.386	2.382	2.446	2.397	2.203	2.090	(41)	
	Ebn at Noon		524	692	793	828	862	870	836	808	775	675	479	403	(42)	
	Edh at Noon		50	80	100	109	106	104	107	100	81	63	47	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.31	0.95	2.38	3.85	5.26	5.41	5.25	4.12	2.58	1.10	0.33	0.19	(44)	
	RadStd		0.06	0.18	0.31	0.47	0.56	0.47	0.54	0.41	0.30	0.17	0.05	0.03	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (26 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	-170	-170	N/A	N/A	

Nomenclature: See separate page